Hall Ticket Number:

MR-22

Code:B324CB2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Regular/Supplementary Examinations, December 2024/January 2025

Discrete Mathematics

(Common to CSB and CSD)

Time: 3 Hours | Max. Marks: 60

Answer ALL Questions

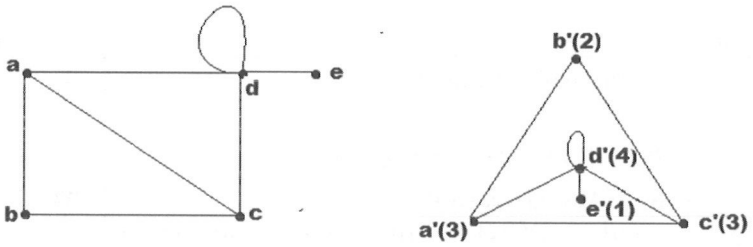
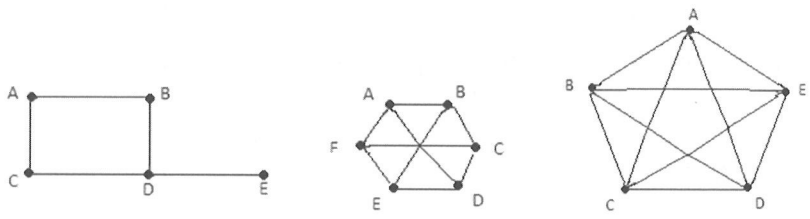
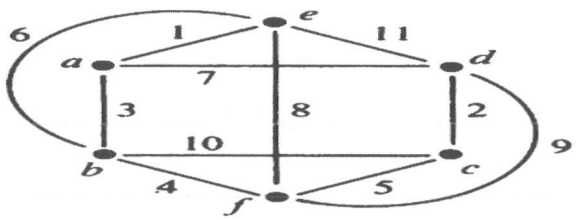
Part-A (10 x 1= 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What is a Well formed formula?	1	1	1	1
1. b)	Translate the sentence **"Not all carshave carburetors."** into symbols, first using no existential quantifier, and second using no universal quantifier.	1	2	1	2
1. c)	Let the Relation R be R= {(1,2) ,(2,3),(3,3)} on the set A= {1,2,3}. What is the Transitive Closure of R?	1	2	2	2
1. d)	Define Equivalence relation	1	1	2	1
1. e)	Define cyclic group with an example.	1	1	3	1
1. f)	Define Monoid.	1	1	3	1
1. g)	How many 4-digit numbers can be formed using the digits2,3,5,6,8, and 9 if repetitions are allowed?	1	2	4	2
1. h)	A farmer buys 3 cows, 8 pigs, and 12chickensfrom a man who has 9 cows,25 pigs, and 100chickens.How many choices does the farmer have?	1	2	4	2
1. i)	Define Hamiltonian path	1	1	5	1
1. j)	Find the chromatic number of bipartite graph $K_{3,3}$?	1	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Obtain the PDNF and PCNF of the statement **PV(¬P→(QV(¬Q→R)))**	5	3	1	2
2. b)	Prove that the following argument is valid: "All dogs are carnivorous. " "Some animals are dogs." Therefore" some animals are carnivorous".	5	5	1	4
	OR				
2. c)	Verify that the following argument is valid by using the rules of inference: If Clifton does not live in France, then he does not speak French. Clifton does not drive a Datsun. If Clifton lives in France, then he rides a bicycle. Either Clifton speaks French, or he drives a Datsun. Hence, Clifton rides a bicycle.	5	4	1	4
2. d)	Show that (¬Q∨(P→ Q)) => ¬P.	5	3	1	2
3. a)	If A={1,2,3,4} and R,S are relations on A defined by R={(1,2),(1,3),(2,4),(4,4)} S={(1,1),(1,2),(1,3),(1,4),(2,3),(2,4)} find R o S, S o R, R^2 , S^2 , write down there matrices.	5	1	2	2
3. b)	What is Compatibility relation and Write the procedure to find maximal compatibility blocks.	5	2	2	1
	OR				
3. c)	If X= {1,2,3,4} and R= {(x,y) /x<y} .Draw the graph of 'R' and also give its matrix.	5	3	2	2

P.T.O.

3. d)	Give an example of relation which is symmetric but neither reflexive nor anti symmetric nor transitive.	5	2	2	1
4. a)	Prove that G={0,1,2,3,4,5,6} is an abelian group of order 7 with respect to addition modulo 7.	5	3	3	2
4. b)	Prove that in a group its identity element, inverse element are unique.	5	3	3	2
	OR				
4. c)	Draw the Hasse diagram representing the partial ordering {(A,B):A subset of B} on the power set P(S) where S={a,b,c}. Find the maximal, minimal, greatest and least elements of the poset.	10	1	3	6
5. a)	A bag of coins contains 10 nickels, 8 dimes, and 7 quarters. Assuming that the coins of any one denomination are indistinguishable, in how many ways can 6 coins be selected from the bag?	5	4	4	2
5. b)	In how many ways can 23 different books be given to 5 students so that 2 of the students will have 4 books each and the other 3 will have 5 books each?	5	4	4	2
	OR				
5. c)	What is the coefficient of x^3y^7 in $(x + y)^{10}$ and in $(2x - 9y)^{10}$	5	3	4	1
5. d)	Determine the coefficient of $x^5y^{10}z^5w^5$ in $(x - 7y + 3z - w)^{25}$.	5	5	4	1
6. a)	Show that the following graphs are isomorphic. 	5	1	5	1
6. b)	Explain Prim's spanning tree algorithm.	5	2	5	2
	OR				
6. c)	Find an Euler path or an Euler Circuit if it exists, in each of the three graphs given below. 	5	1	5	1
6. d)	Find a minimal spanning tree for the below graph. 	5	3	5	2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome